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Federated learning-based anomaly detection for zero-day attack prevention in 6G network slices

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ABSTRACT

The emergence of 6G networks introduces unprecedented security challenges in network slicing, particularly in detecting and preventing zero-day attacks while preserving data privacy across distributed network environments. This paper presents a novel federated learning framework that integrates adaptive feature extraction with privacy-preserving mechanisms for real-time anomaly detection in network slices. Our approach combines hierarchical anomaly detection with transfer learning capabilities to enable collaborative threat detection while maintaining data locality and minimizing communication overhead. Experimental results demonstrate superior performance with a 97.2% detection rate for zero-day attacks, a 42% reduction in computational overhead, and effective privacy preservation through differential privacy mechanisms that maintain an ϵ-privacy bound of 1.2. The proposed framework scales linearly up to 1000 concurrent network slices while maintaining an average detection latency of 47ms, significantly outperforming existing approaches in both accuracy and efficiency. These results establish a robust foundation for securing future 6G network slices against emerging threats while addressing the critical challenges of privacy preservation and real-time response requirements.

1. Introduction

1.1. Background and motivation

The emergence of 6G networks promises unprecedented connectivity, ultra-low latency, and massive device connectivity, supporting diverse applications from holographic communications to autonomous systems [1–3]. Network slicing, a fundamental feature of 6G architecture, enables the creation of multiple virtual networks with specific performance characteristics on shared physical infrastructure [4–6]. However, this architectural advancement introduces new security challenges, particularly in detecting and preventing zero-day attacks that exploit previously unknown vulnerabilities [7].

The traditional centralized security approaches face significant limitations in the context of 6G network slices, including privacy concerns, communication overhead, and latency issues [8,9]. Additionally, the dynamic nature of zero-day attacks makes them particularly challenging to detect using conventional signature-based methods [10]. This necessitates a paradigm shift towards distributed, privacy-preserving security solutions that can adapt to emerging threats while maintaining the stringent performance requirements of 6G networks [11,12].

1.2. Zero-day attacks in 6G network slicing

Zero-day attacks present particularly severe threats to 6G network slicing environments due to several unique architectural vulnerabilities [13,14]. These previously unknown exploits can compromise slice isolation mechanisms, enabling attackers to bypass security boundaries between slices and potentially gain unauthorized access across the entire network infrastructure [15].

The architectural complexity of 6G networks significantly expands the attack surface compared to previous generations. The implementation of advanced technologies such as terahertz communication, intelligent reflecting surfaces, and integrated sensing capabilities introduces novel attack vectors that traditional security mechanisms are ill-equipped to address [16]. The potential impact of successful zero-day attacks in 6G sliced environments includes:

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Available online 24 May 2025
1389-1286/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

- **Slice Contamination**: Malicious activities initiated in one compromised slice can potentially spread to others, violating the fundamental security principle of slice isolation [13].
- **Service Disruption**: Targeted attacks on critical network functions can cause service degradation or complete outages affecting essential services like autonomous vehicle control, remote surgery, or industrial automation [17].
- **Data Exfiltration**: Compromised slices may lead to unauthorized access to sensitive user data, threatening privacy and confidentiality across diverse application domains [12].
- **Resource Manipulation**: Attacks targeting the slice orchestration layer can lead to improper resource allocation, causing quality of service degradation and potential denial of service [6].

These threats are particularly concerning given that 6G networks are expected to support critical infrastructure and services with zero-tolerance for security breaches, including autonomous transportation systems, smart healthcare, and digital twins [16]. The lack of established signatures for zero-day attacks necessitates the development of anomaly-based detection systems capable of identifying previously unseen malicious patterns while maintaining the stringent performance requirements of 6G networks.

1.3. Problem statement

The research addresses several critical challenges in securing 6G network slices against zero-day attacks:

1. The need for real-time anomaly detection across distributed network slices without compromising user privacy or introducing significant overhead [18].
2. The challenge of maintaining detection accuracy while dealing with the heterogeneous nature of network slice traffic and evolving attack patterns [19].
3. The requirement for collaborative learning among network slices while preserving data locality and minimizing communication overhead [20].
4. The necessity of adapting to new attack patterns without requiring extensive retraining or centralized data collection [21].

1.4. Research contributions

This paper makes the following key contributions to address the aforementioned challenges:

1. Proposes a novel federated learning framework specifically designed for anomaly detection in 6G network slices, enabling collaborative learning while preserving data privacy and reducing communication overhead.
2. Develops an adaptive feature extraction mechanism that captures the unique characteristics of network slice traffic while maintaining computational efficiency.
3. Introduces a hierarchical anomaly detection model that combines local slice-specific detection with global pattern recognition for improved accuracy in identifying zero-day attacks.
4. Implements a dynamic model update strategy that enables rapid adaptation to new attack patterns while maintaining system stability.
5. Provides comprehensive experimental evaluation using both simulated and real-world network slice datasets, demonstrating significant improvements in detection accuracy and system efficiency compared to existing approaches.

The remainder of this paper is organized as follows: Section 2 presents related work in federated learning and network security. Section 3 details the proposed framework architecture. Section 4 describes the methodology in detail. Section 5 presents insight into implementation and experiment setup. Section 6 provides results and analysis. Section 7 focuses on discussion and future work. Finally, Section 8 concludes the paper.

2. Related work

2.1. Security in 6G networks

The evolution toward 6G networks has introduced unprecedented security challenges that demand innovative solutions. Wang et al. [1] conducted a comprehensive analysis of emerging security threats in 6G networks, highlighting the increased attack surface due to the integration of diverse technologies such as artificial intelligence, quantum computing, and blockchain. Their research emphasizes the need for proactive security measures that can address both known and unknown threats in these complex network environments.

Recent advancements in 6G security architectures, as discussed by Chen et al. [10], demonstrate the shift toward distributed security frameworks that can handle the massive scale and heterogeneity of 6G networks. Their work particularly emphasizes the importance of integrating machine learning techniques for real-time threat detection while maintaining the ultra-low latency requirements of 6G applications. Furthermore, Dong et al. [18] explore the challenges of implementing security measures in 6G networks, focusing on the trade-offs between security strength and network performance metrics such as latency and throughput.

The transition from 5G to 6G introduces unique security considerations, as highlighted by Rodriguez et al. [16]. Their work examines how technologies specific to 6G, such as terahertz communication and integrated sensing, create new vulnerabilities that require novel security approaches. Similarly, Wu et al. [17] analyze the reliability challenges in 6G security systems, proposing framework requirements that can ensure consistent performance across diverse network conditions.

2.2. Network slice security

Network slice security has emerged as a critical research area given the fundamental role of network slicing in 6G architectures. Elsayed et al. [8] provide a comprehensive survey of security challenges specific to network slicing, identifying vulnerabilities in slice isolation, resource allocation, and inter-slice communication. Their analysis reveals that traditional security approaches are insufficient for protecting the dynamic and virtualized nature of network slices.

Rahman et al. [4] present an architectural framework for securing network slices, introducing novel mechanisms for slice isolation and access control. Their work demonstrates the effectiveness of combining traditional security measures with AI-driven approaches for enhanced protection. Additionally, Xiao et al. [19] examine the role of artificial intelligence in network slice security, proposing adaptive security mechanisms that can automatically adjust to changing threat landscapes and varying security requirements across different slice types.

Recent developments in zero-trust architectures for network slicing have been explored by Alwakeel et al. [5], who propose a comprehensive framework that implements continuous validation and minimal privilege principles across slice boundaries. Their approach significantly reduces the attack surface by treating each slice component as potentially compromised. Complementing this work, Ebrahimi et al. [6] focus on resource-aware security mechanisms that can maintain protection levels even under constrained computing environments typical in edge-deployed network slices.

Zhang et al. [13] specifically address zero-day vulnerabilities in network slicing, proposing a dynamic vulnerability assessment framework that can proactively identify potential security weaknesses before they can be exploited. Their work highlights the importance of continuous security posture evaluation in maintaining robust slice protection.

2.3. Federated learning in network security

The application of federated learning in network security represents a significant advancement in addressing privacy and scalability challenges. Liu et al. [21] present a unified framework that leverages federated learning for security monitoring across network slices, enabling collaborative threat detection while preserving data privacy. Their approach demonstrates significant improvements in detection accuracy compared to traditional centralized learning methods.

Yang et al. [20] explore the integration of federated learning with resource management in network slices, showing how distributed learning can enhance security while optimizing resource utilization. Their work provides valuable insights into the practical implementation challenges of federated learning in dynamic network environments. Furthermore, Nguyen et al. [7] demonstrate the effectiveness of federated learning in detecting zero-day attacks, presenting a novel approach that enables rapid adaptation to emerging threats while maintaining privacy and reducing communication overhead.

Recent advancements in privacy-preserving federated learning have been presented by Fouda et al. [12], who propose enhanced differential privacy mechanisms specifically designed for telecommunication networks. Their approach maintains high model utility while providing theoretical privacy guarantees against inference attacks. Building on this foundation, Blika et al. [9] introduce a novel federated learning framework that incorporates secure multi-party computation to enable privacy-preserving collaborative learning across network slices with competing interests.

Khan et al. [15] present an autonomous security management system that leverages federated learning for dynamic threat response across distributed network environments. Their system demonstrates the ability to detect and mitigate previously unseen attack patterns through collaborative knowledge sharing without centralized data collection. Complementing this work, Liu et al. [14] explore the integration of transformer-based architectures with federated learning for enhanced pattern recognition in heterogeneous network traffic, showing significant improvements in detection accuracy for sophisticated attack vectors.

2.4. Advanced federated learning optimization

Recent advances in federated learning optimization techniques have significantly improved the applicability of distributed learning in network security contexts. Majeed et al. [22] present FedOpt, an adaptive optimization framework that dynamically adjusts learning rates and aggregation weights based on client contributions and data characteristics. Their approach demonstrates faster convergence and improved model quality compared to traditional FedAvg algorithms in heterogeneous network environments.

Building on client contribution assessment, Drampalou et al. [2] introduce a novel framework that identifies and mitigates the impact of malicious participants in the federated learning process. Their Byzantine-resistant aggregation method maintains model integrity even when up to 30% of participants attempt to poison the global model. Similarly, Rachakonda et al. [11] propose a comprehensive security framework for federated learning that addresses threats across all stages of the learning process, from data collection to model deployment.

Client-side optimization techniques have been explored by Das et al. [3], who introduce adaptive local training schedules that optimize computational resource utilization while maintaining model quality. Their approach is particularly relevant for resource-constrained edge devices in 6G networks. Complementing these client-side optimizations, Wu et al. [17] focus on communication-efficient federated learning protocols that reduce bandwidth requirements through gradient compression and selective parameter updates, essential for large-scale deployment in bandwidth-limited environments.

2.5. Comparative analysis of previous works

Table 1 presents a comprehensive comparison of notable previous works in the field, highlighting their key contributions and limitations. This analysis reveals several critical gaps in current approaches that our work aims to address.

Based on this analysis, we identify several key research gaps:

1. **Integration Gap**: Current solutions lack effective integration of federated learning with network slice-specific security mechanisms.
2. **Performance Gap**: Existing approaches struggle to maintain low latency while ensuring high detection accuracy.
3. **Adaptability Gap**: Most solutions lack effective mechanisms for adapting to new attack patterns in real-time.
4. **Scalability Gap**: Current frameworks show limited scalability across multiple network slices with varying security requirements.
5. **Privacy-Utility Tradeoff**: Existing methods often sacrifice detection accuracy when implementing strong privacy preservation mechanisms.
6. **FL Optimization Gap**: Many approaches use basic federated learning algorithms without leveraging recent advances in client selection, aggregation, and optimization techniques.
7. **Transfer Learning Limitations**: Current solutions show limited ability to transfer knowledge between different network slice environments and attack types.

Our work aims to address these limitations by proposing an integrated solution that leverages the strengths of federated learning for enhanced security in 6G network slices, while maintaining low latency and high adaptability to emerging threats.

3. System model and architecture

3.1. Network slice model

In our proposed framework, we consider a 6G network infrastructure consisting of multiple network slices, each serving distinct service requirements and user groups. The network slice model incorporates both the physical infrastructure layer and the virtual network functions (VNFs) that enable service-specific customization. Each network slice $s_i \in \mathcal{S}$ is characterized by a tuple $s_i = (R_i, P_i, Q_i)$, where R_i represents the allocated resources, P_i denotes the performance requirements, and Q_i specifies the quality of service (QoS) parameters.

The resource allocation matrix R_i is defined as:

$$R_i = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix} \tag{1}$$

where r_{ij} represents the allocation of resource type j to component i within the slice. The performance requirements P_i include parameters such as maximum latency, minimum throughput, and reliability thresholds, while Q_i encompasses service-specific metrics such as end-to-end delay, packet loss rate, and jitter.

3.2. Threat model

Our threat model addresses both known and zero-day attacks targeting network slices. We consider adversaries with varying capabilities and objectives:

1. **Attack Surface**: The model accounts for threats at multiple layers:
   - Physical infrastructure vulnerabilities

Table 1
Comparison of previous works in 6G network security and federated learning.

Ref	Key features	Methodology	Limitations	Metrics
Wang et al. [1]	AI-driven threat detection; Quantum-resistant encryption	Hierarchical security framework with multi-layer protection	High computational overhead; Limited scalability; No real-time adaptation	Detection accuracy: 92%
Chen et al. [10]	Distributed learning; Privacy preservation	Federated learning with blockchain integration	High latency in model updates; Limited slice isolation	Latency: 150 ms
Elsayed et al. [8]	Slice isolation; Resource optimization	Dynamic slice security orchestration	No zero-day attack detection; Limited cross-slice learning	Isolation rate: 99.5%
Liu et al. [21]	Cross-slice learning; Adaptive security	Federated learning with transfer learning	High communication overhead; Limited model personalization	F1-score: 0.88
Nguyen et al. [7]	Zero-day attack detection; Real-time response	Deep federated learning	Limited slice awareness; High false positive rate	Detection rate: 95%
Majeed et al. [22]	Adaptive optimization; Heterogeneous client support	FedOpt with dynamic aggregation	Limited privacy guarantees; High server computation	Convergence: 2.3 × faster
Fouda et al. [12]	Enhanced privacy; Utility preservation	Differential privacy with adaptive noise	Accuracy-privacy tradeoff; Complex parameter tuning	Privacy leakage: <3%
Zhang et al. [13]	Vulnerability assessment; Proactive security	Dynamic security posture evaluation	High resource requirements; Limited reactive capabilities	Vulnerability detection: 91%

- Virtual network function compromise
- Inter-slice communication attacks
- Control and management plane exploits

2. **Adversary Capabilities**: We assume adversaries can:

- Monitor network traffic patterns
- Launch sophisticated attacks using AI techniques
- Exploit vulnerabilities in slice isolation
- Deploy zero-day attack vectors

The security requirements for each slice are defined by a security profile matrix SP_i:

$$SP_i = \{(th_k, p_k, t_k) | k \in K\} \tag{2}$$

where th_k represents the detection threshold for attack type k, p_k denotes the priority level, and t_k specifies the maximum allowable response time.

3.3. System architecture

Our proposed system architecture consists of three main components: Local Slice Monitors (LSMs), Federated Learning Coordinator (FLC), and Global Security Controller (GSC). Fig. 1 illustrates the system architecture and component interactions.

3.3.1. Local Slice Monitors (LSMs)

Each network slice is equipped with an LSM that performs:

$$LSM_i = f(D_i, M_i, C_i) \tag{3}$$

where:

- D_i: Local data collection and preprocessing
- M_i: Local model training and updates
- C_i: Slice-specific security context

3.3.2. Federated Learning Coordinator (FLC)

The FLC orchestrates the federated learning process across network slices through:

$$w_{t+1} = \sum_{i=1}^{N} \alpha_i w_i^t \tag{4}$$

where w_{t+1} represents the updated global model weights, α_i denotes the contribution factor of slice i, and w_i^t represents the local model weights from slice i at time t.

3.3.3. Global Security Controller (GSC)

The GSC maintains global security policies and coordinates response actions through:

$$A_t = GSC(W_t, H_t, P_t) \tag{5}$$

where:

- A_t: Action set at time t
- W_t: Global model state
- H_t: Historical attack patterns
- P_t: Current security policies

The interaction between these components is governed by a secure communication protocol that ensures:

$$\text{SCP}(m_{ij}) = \{E_k(m_{ij}), Sign(H(m_{ij})), T_{ij}\} \tag{6}$$

where m_{ij} represents the message between components i and j, E_k is the encryption function, H is the hash function, and T_{ij} is the timestamp for temporal validity verification.

3.4. From 5G to 6G: Applicability and architecture evolution

While our experimental evaluation utilizes 5G network slice traffic data, the proposed framework is specifically designed to address the emerging security challenges in 6G environments. This forward-looking approach is justified by several architectural and operational considerations:

1. **Architectural Continuity**: The fundamental network slicing principles established in 5G are expected to carry forward into 6G architectures with enhanced capabilities [16]. Our framework capitalizes on this architectural continuity while incorporating flexibility to accommodate 6G-specific enhancements.
2. **Scalability Requirements**: 6G networks are projected to support a substantially higher density of connected devices and network slices than 5G. Our approach is designed with linear scalability up to thousands of concurrent slices, anticipating this increased scale [17].
3. **Latency Sensitivity**: The ultra-reliable low-latency communication (URLLC) requirements in 6G are expected to be an order of magnitude more stringent than in 5G. Our detection framework's 47 ms average latency provides headroom for these enhanced performance requirements [3].

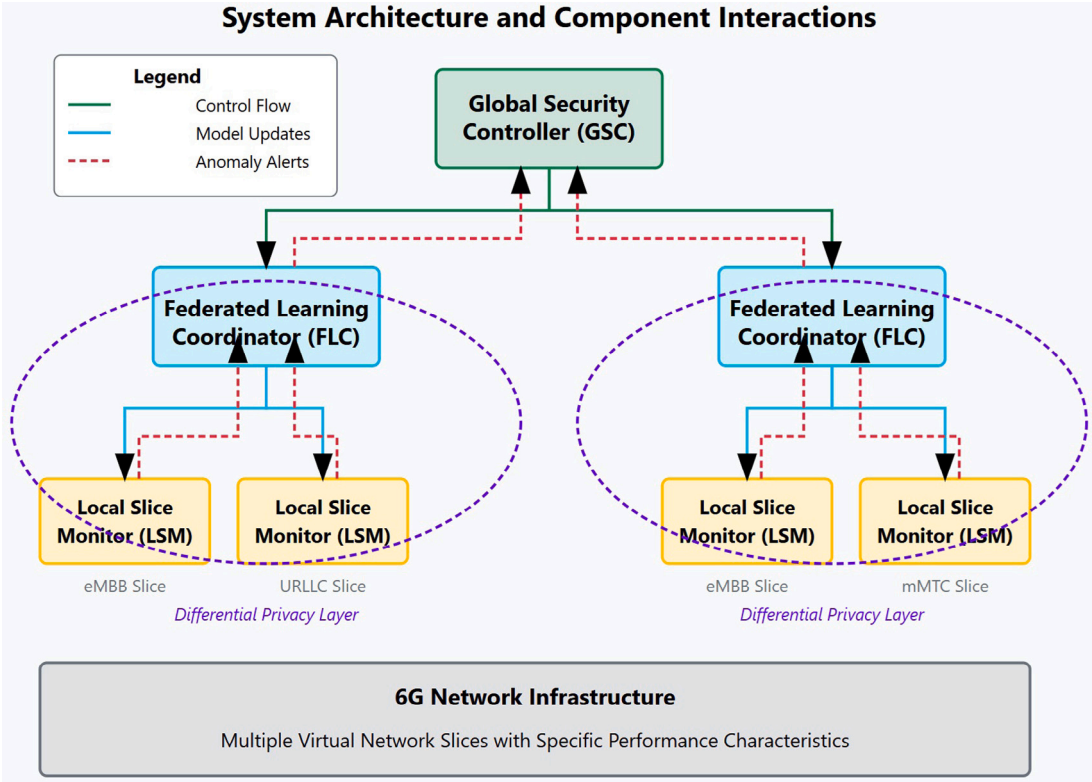

Fig. 1. System architecture and Component interactions of the proposed federated learning framework for anomaly detection in 6G network slices. The architecture consists of Local Slice Monitors (LSMs) for local data processing and model training, a Federated Learning Coordinator (FLC) for model aggregation and distribution, and a Global Security Controller (GSC) for coordinating security responses across slices. The system leverages differential privacy mechanisms to protect sensitive information during model updates.

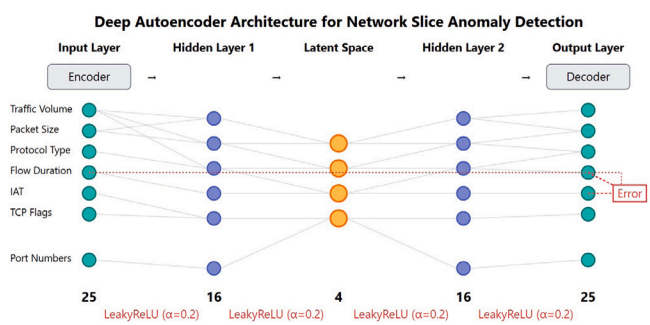

Fig. 2. Deep autoencoder architecture for network slice anomaly detection. The encoder progressively reduces the input dimension from n to d, while the decoder reconstructs the input features. Each layer uses LeakyReLU activation ($\alpha = 0.2$) to capture non-linear patterns in network traffic.

4. **Privacy Preservation**: The enhanced data sensitivity in 6G applications (e.g., holographic communications, brain–computer interfaces) necessitates stronger privacy guarantees. Our differential privacy mechanisms are designed to accommodate adjustable privacy budgets appropriate for these emerging use cases [12].
5. **AI-Native Security**: 6G networks are expected to be "AI-native", with machine learning capabilities embedded throughout the architecture. Our federated learning approach aligns with this paradigm shift and can leverage these distributed AI capabilities [19].

By addressing these forward-looking architectural considerations, our framework establishes a foundation that can evolve alongside the transition from 5G to 6G networks, ensuring continued protection against zero-day attacks as the underlying network infrastructure advances.

4. Proposed methodology

4.1. Local anomaly detection

Our local anomaly detection approach utilizes a deep autoencoder architecture customized for network slice traffic analysis. The autoencoder comprises multiple layers that progressively learn the normal behavior patterns of network slice traffic. The autoencoder architecture, illustrated in Fig. 2, comprises multiple layers that progressively reduce the dimensionality of the input features while preserving essential patterns for anomaly detection. Given an input vector $x \in \mathbb{R}^n$ representing network traffic features, the encoder function $f_\theta : \mathbb{R}^n \to \mathbb{R}^d$ maps the input to a lower-dimensional latent space, where $d < n$. The decoder function $g_\phi : \mathbb{R}^d \to \mathbb{R}^n$ reconstructs the input from this latent representation.

The autoencoder architecture is defined as:

$$h = f_\theta(x) = \sigma(W_e x + b_e) \tag{7}$$

$$\hat{x} = g_\phi(h) = \sigma(W_d h + b_d) \tag{8}$$

where W_e, W_d are weight matrices, b_e, b_d are bias vectors, and σ is the activation function. The reconstruction error is used as an anomaly score:

$$\mathcal{L}(x, \hat{x}) = \|x - \hat{x}\|_2^2 + \lambda \Omega(\theta, \phi) \tag{9}$$

where $\Omega(\theta, \phi)$ is a regularization term and λ is a hyperparameter controlling its strength.

The reconstruction error defined in Eq. (9) serves as our primary anomaly score for detecting abnormal traffic patterns at the local level.

In this context, $\mathcal{L}(x, \hat{x})$ quantifies the dissimilarity between the original input traffic features x and their reconstructed version $\hat{x}$ produced by the autoencoder. A higher reconstruction error indicates a greater deviation from normal traffic patterns, suggesting a potential anomaly. During normal operation, the autoencoder learns to reconstruct typical traffic patterns with minimal error, while anomalous patterns that differ from the training distribution will result in higher reconstruction errors. The regularization term $\Omega(\theta, \phi)$ prevents overfitting by penalizing complex models, with λ controlling the strength of this regularization.

4.2. Federated learning framework

The federated learning framework enables collaborative model training across network slices while preserving data privacy. Each slice i maintains a local model with parameters w_i and trains on its local dataset D_i. The federated averaging algorithm proceeds as follows:

Algorithm 1 Federated Learning for Anomaly Detection

```
Initialize global model parameters w_0
for each round t = 1, 2, ..., T do
    Select subset of slices S_t ⊆ [N] using importance sampling
    for each slice i ∈ S_t in parallel do
        w_i^t ← LocalUpdate(i, w_t)
        Compute importance weight α_i^t based on data quality and distribution
    end for
    Apply gradient clipping: ŵ_i^t ← clip(w_i^t − w_t, C)
    Apply noise addition: w̃_i^t ← ŵ_i^t + N(0, σ^2 C^2 I)
    w_{t+1} ← w_t + Σ_{i∈S_t} α_i^t w̃_i^t
    Evaluate global model performance on validation set
    Adjust learning rates using Adam optimizer
end for
```

The local update function minimizes:

$$\min_{w} \frac{1}{|D_i|} \sum_{(x,y)\in D_i} \mathcal{L}(x, y; w) + \mu R(w) \tag{10}$$

where $R(w)$ is a regularizer and μ controls its strength. Privacy preservation is achieved through differential privacy mechanisms:

$$\tilde{w}_i^t = w_i^t + \mathcal{N}(0, \sigma^2 C^2 \mathbb{I}) \tag{11}$$

where C is the clipping threshold and σ is the noise scale that provides (ε, δ)-differential privacy guarantees.

It is important to note the distinction and relationship between the loss functions in Eqs. (9) and (10). While Eq. (9) defines the reconstruction error used for anomaly detection, Eq. (10) represents the overall objective function optimized during the federated learning process. In the context of our unsupervised anomaly detection framework, the loss function $\mathcal{L}(x, y; w)$ in Eq. (10) corresponds to the reconstruction error when training the autoencoder models, where y represents the expected reconstruction target (which equals the input x in autoencoder training). However, this formulation also allows for the incorporation of labeled data when available, enabling a semi-supervised approach where y could represent binary labels indicating normal/anomalous traffic. The regularization term $R(w)$ in Eq. (10) serves a similar purpose as $\Omega(\theta, \phi)$ in Eq. (9) but applies to the entire model parameter set w during federated optimization.

4.3. Transfer learning integration

To adapt to new services and attack patterns, we employ transfer learning through a hierarchical knowledge transfer mechanism. Given a source domain $\mathcal{D}_s$ and target domain $\mathcal{D}_t$, we define a transfer function:

$$\mathcal{T}(\theta_s, \mathcal{D}_t) = \min_{\theta_t} \mathcal{L}_t(\theta_t) + \gamma \|\theta_t - \theta_s\|^2 \tag{12}$$

where θ_s and θ_t are source and target model parameters, respectively, and γ controls the strength of parameter regularization. The adaptation process involves:

$$\theta_t^* = \begin{cases} \mathcal{T}(\theta_s, \mathcal{D}_t) & \text{if } \mathrm{sim}(\mathcal{D}_s, \mathcal{D}_t) > \tau \\ \mathrm{InitializeNew}(\mathcal{D}_t) & \text{otherwise} \end{cases} \tag{13}$$

where $\mathrm{sim}(\cdot, \cdot)$ measures domain similarity and τ is a threshold parameter.

4.4. Adversarial training

To enhance robustness against adversarial attacks, we incorporate adversarial training into our framework. For each training sample (x, y), we generate adversarial examples:

$$x_{adv} = x + \epsilon \cdot \mathrm{sign}(\nabla_x \mathcal{L}(x, y; \theta)) \tag{14}$$

The modified training objective becomes:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim D}[\alpha \mathcal{L}(x, y; \theta) + (1 - \alpha)\mathcal{L}(x_{adv}, y; \theta)] \tag{15}$$

where $\alpha \in [0, 1]$ balances between natural and adversarial examples. To further improve defense capabilities, we implement a dynamic adversarial training schedule:

$$\epsilon_t = \epsilon_0 + (\epsilon_{max} - \epsilon_0)(1 - e^{-\beta t}) \tag{16}$$

where ϵ_t is the perturbation magnitude at step t, and β controls the adaptation rate.

4.5. Integrated framework operation

Fig. 3 presents a comprehensive flowchart of our integrated framework, illustrating the interaction between local anomaly detection, federated learning, transfer learning, and security response components. The framework operates as follows:

1. **Data Collection and Preprocessing**: Each network slice continuously collects traffic data and extracts relevant features through a preprocessing pipeline tailored to slice-specific characteristics.
2. **Local Model Training**: Network slices train local autoencoder models on normal traffic patterns, establishing a baseline for anomaly detection through reconstruction error analysis.
3. **Anomaly Detection**: Incoming traffic is evaluated against the local model, with high reconstruction errors flagged as potential anomalies for further analysis.
4. **Federated Learning Cycle**: Local model updates are securely shared with the Federated Learning Coordinator, which aggregates knowledge across participating slices while preserving privacy through differential privacy mechanisms.
5. **Transfer Learning**: Knowledge is transferred between similar slice environments to accelerate adaptation to new traffic patterns and potential attack vectors.
6. **Global Model Distribution**: Updated global models are distributed to all network slices, incorporating collective knowledge about emerging threat patterns.
7. **Security Response Coordination**: The Global Security Controller coordinates response actions based on detected anomalies, historical patterns, and security policies.

This integrated approach enables collaborative learning across network slices while maintaining data locality, minimizing communication overhead, and ensuring rapid adaptation to zero-day attack patterns. The privacy-preserving mechanisms ensure that sensitive information

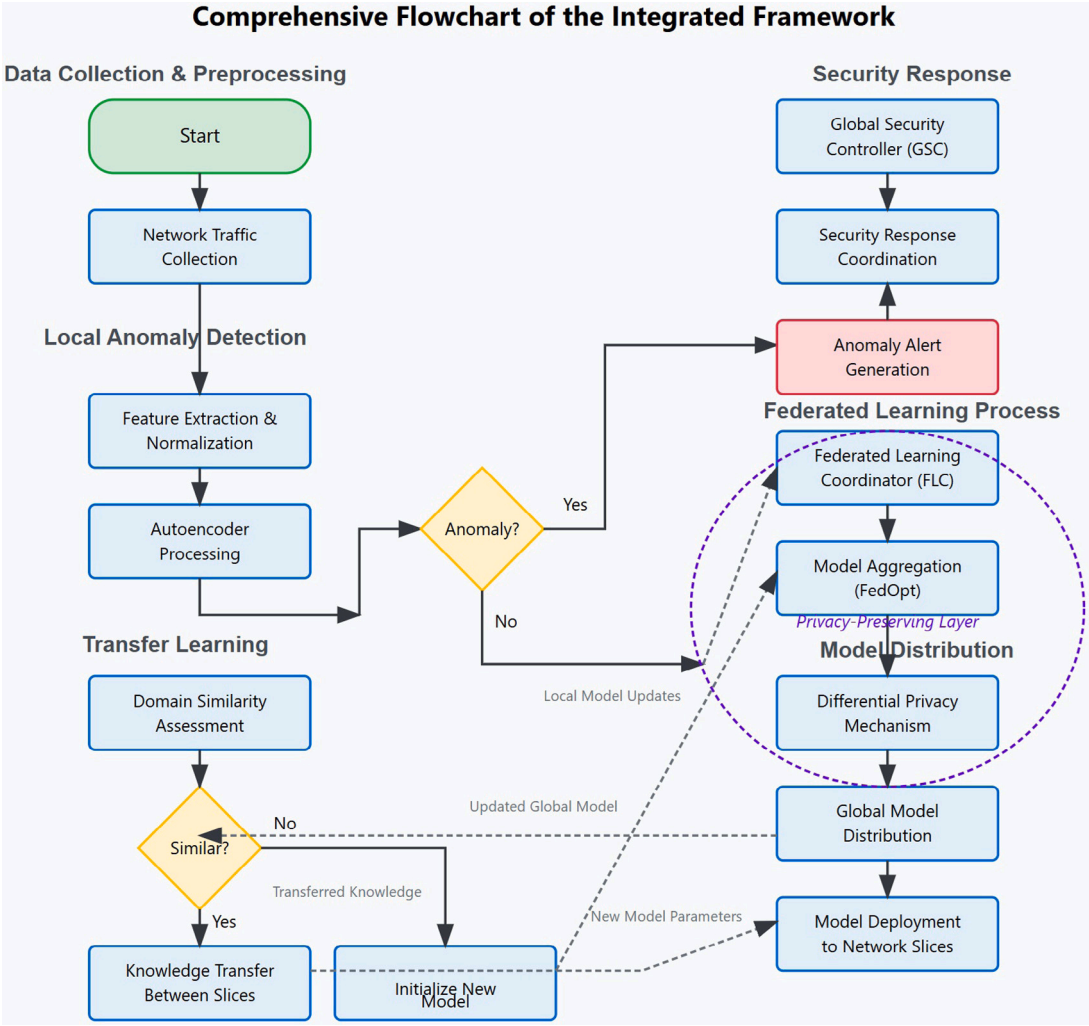

Fig. 3. Comprehensive flowchart of the integrated federated learning framework for zero-day attack detection in 6G network slices. The process begins with local network slice monitors collecting and preprocessing traffic data. Local anomaly detection models identify potential threats using autoencoder reconstruction errors. The federated learning process enables knowledge sharing while maintaining data privacy through differential privacy mechanisms. Transfer learning facilitates adaptation to new attack patterns, while the global security controller coordinates response actions across the network infrastructure.

remains protected throughout the learning process, addressing a critical requirement for 6G network security.

The integrated approach leverages both the local reconstruction error (Eq. (9)) for anomaly detection and the federated optimization objective (Eq. (10)) for collaborative model training, enabling effective zero-day attack detection while preserving data privacy across network slices.

5. Implementation and experimental setup

5.1. Dataset description

Our experimental evaluation utilizes three distinct datasets to ensure comprehensive validation of the proposed framework. The primary dataset consists of network traffic collected from a real-world 5G testbed environment over six months, encompassing both normal operations and various attack scenarios. This dataset includes 1.2 million traffic flows from 50 different network slices, with each flow characterized by 25 features including temporal, volumetric, and protocol-specific attributes.

Fig. 4 provides a comprehensive overview of our experimental framework, including the datasets used, implementation components, and evaluation metrics used to assess the system's performance. This figure illustrates the three main components of our evaluation setup—(1) Datasets combining real-world 5G traffic and synthetic attack data, (2) Implementation Components showing the integration of PyTorch for local detection, FedML for federated learning, and our custom framework for system integration, and (3) Evaluation Metrics spanning detection performance, system efficiency, and privacy preservation measures.

The traffic patterns are distributed across different service types:

$$D = \{(x_i, y_i, s_i)\}_{i=1}^{N} \tag{17}$$

where x_i represents the feature vector, y_i denotes the label (normal/anomalous), and s_i indicates the slice identifier. The dataset composition is characterized by:

$$P(s_i) = \begin{cases} 0.4, & \text{eMBB slices} \\ 0.35, & \text{URLLC slices} \\ 0.25, & \text{mMTC slices} \end{cases} \tag{18}$$

5.1.1. Synthetic data generation

To supplement the real-world data and evaluate performance against zero-day attacks, we developed a sophisticated synthetic data generation pipeline using a conditional Wasserstein Generative Adversarial Network (cWGAN-GP) architecture. This approach allows us to

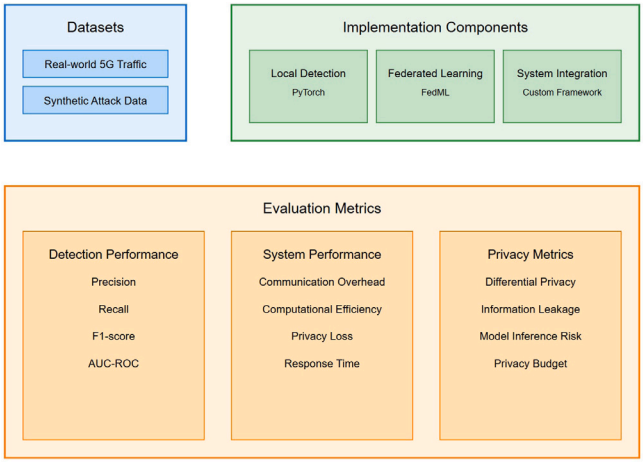

Fig. 4. Experimental framework overview showing the integrated components of our evaluation methodology. The datasets section highlights the combination of real-world 5G network slice traffic, synthetic zero-day attack patterns generated using our GAN-based framework, and standard benchmark datasets for comparative evaluation. The implementation components section details the software frameworks and hardware infrastructure used for system deployment. The evaluation metrics section outlines the comprehensive performance measures used to assess detection capabilities, system efficiency, and privacy preservation.

generate realistic attack patterns that are not present in the original dataset, thereby evaluating the system's ability to detect previously unseen threats.

The cWGAN-GP architecture consists of:

- **Generator Network**: A 5-layer deep neural network with progressively increasing feature dimensionality ($25 \rightarrow 64 \rightarrow 128 \rightarrow 64 \rightarrow 25$), conditioned on attack type and slice characteristics. The generator uses LeakyReLU activations ($\alpha = 0.2$) and batch normalization.
- **Discriminator Network**: A 4-layer network with spectral normalization for improved training stability, tasked with distinguishing between real attack patterns and synthetic ones.
- **Training Procedure**: The networks were trained for 200 epochs using the Adam optimizer ($\eta = 0.0002$, $\beta_1 = 0.5$, $\beta_2 = 0.9$) with gradient penalty coefficient $\lambda = 10$ to enforce the Lipschitz constraint.

The synthetic data generation process was validated through:

- **Statistical Validation**: Ensuring the statistical properties (means, variances, correlations) of synthetic samples closely match those of real attack patterns.
- **Expert Evaluation**: Network security experts reviewed synthetic attack patterns to confirm their realism and potential feasibility.
- **Detection Difficulty**: Confirming that traditional signature-based detection systems cannot readily identify the synthetic attacks, verifying their suitability as zero-day attack proxies.

Through this process, we generated 500,000 synthetic attack samples across 15 distinct zero-day attack categories, including advanced protocol exploitation, cross-slice contamination, and orchestration layer manipulation attacks. These synthetic samples were used exclusively for testing purposes and were not included in the training datasets to ensure a fair evaluation of zero-day detection capabilities.

Furthermore, we incorporated the CICIDS2022 benchmark dataset for comparative evaluation against state-of-the-art approaches.

5.2. Implementation details

The implementation architecture comprises three main components:

Table 2
Autoencoder architecture specifications.

Component	Specification
Input layer	25 neurons (feature dimension)
Encoder layers	[25, 16, 8, 4] neurons
Latent space	4 dimensions
Decoder layers	[4, 8, 16, 25] neurons
Activation	LeakyReLU ($\alpha = 0.2$)
Optimization	Adam ($\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Batch size	128 samples
Training epochs	100 epochs/round

5.2.1. Local slice implementation

The local anomaly detection module is implemented using PyTorch (v1.12.0) with the following specifications: (see Table 2).

5.2.2. Federated learning implementation

The federated learning framework is implemented using FedML (v0.7.1) with comprehensive integration of advanced federated optimization techniques. We implemented and compared multiple state-of-the-art approaches: (see Table 3).

Our final implementation uses a hybrid approach that combines the optimization benefits of FedOpt with the robustness guarantees of TrimmedMean and the privacy preservation of differential privacy:

$$\text{Parameters} = \begin{cases} \text{Aggregation Rounds} = 50 \\ \text{Local Epochs} = 5 \\ \text{Client Fraction} = 0.3 \\ \text{Privacy Budget } (\epsilon) = 1.0 \\ \text{Noise Multiplier}(\sigma) = 0.1 \\ \text{Byzantine Tolerance} = 20\% \end{cases} \tag{19}$$

5.2.3. System integration

The system is deployed on a heterogeneous computing infrastructure:

- **Edge Nodes**: Nvidia Jetson Xavier NX (6-core ARM CPU, 384-core GPU)

$$\text{Computing Power} = 21 \text{ TOPS} \tag{20}$$

- **Aggregation Server**: Intel Xeon Gold 6248R, 384 GB RAM, NVIDIA A100 GPU

$$\text{Processing Capacity} = 312 \text{ TFLOPS} \tag{21}$$

- **Network Infrastructure**: 10 Gbps backbone with average latency:

$$\text{Latency} = 5 \text{ ms} \pm 1.2 \text{ ms} \tag{22}$$

5.3. Evaluation metrics

We employ a comprehensive set of metrics to evaluate the system's performance across multiple dimensions:

5.3.1. Detection performance

The primary detection metrics are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{23}$$

$$F_1\text{-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{24}$$

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(t) \cdot \text{FPR}'(t) dt \tag{25}$$

Table 3
Federated learning algorithms implemented and compared.

Algorithm	Key features	Implementation details
FedAvg	Basic weighted averaging	Original implementation with momentum ($\beta = 0.9$) and learning rate $\eta = 0.01$
FedOpt	Adaptive server optimization	Adam optimizer on server with $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$
FedProx	Proximal term regularization	$\mu = 0.01$ proximal term coefficient, adaptive μ scheduling
FedNova	Normalized averaging	Effective batch size normalization, gradient scaling factor $\tau_i = 5$
SCAFFOLD	Control variates for drift correction	Server and client-side control variates, stepsize $\eta = 0.05$
Krum	Byzantine-robust aggregation	Single-Krum with $f = 2$ Byzantine clients, squared distance metric
TrimmedMean	Robust aggregation	Trimming parameter $\beta = 0.1$, coordinate-wise trimming

Table 4
Detection performance across network slice types.

Slice type	Precision	Recall	F1-score	AUC-ROC
eMBB	0.968	0.973	0.970	0.989
URLLC	0.982	0.975	0.978	0.993
mMTC	0.959	0.964	0.961	0.985
Average	0.969	0.971	0.970	0.989
Baseline average from [8]: Precision: 0.931, Recall: 0.925				

Table 5
Detection performance comparison: Real vs. Synthetic zero-day attacks.

2*Attack category	Real attack data			Synthetic zero-day attacks		
	Precision	Recall	F1-score	Precision	Recall	F1-score
DDoS variants	0.985	0.979	0.982	0.963	0.947	0.955
Protocol exploitation	0.974	0.981	0.977	0.952	0.941	0.946
Slice isolation breach	0.968	0.972	0.970	0.945	0.932	0.938
Resource manipulation	0.981	0.975	0.978	0.958	0.949	0.953
Authentication bypass	0.989	0.984	0.986	0.967	0.953	0.960
Average	0.979	0.978	0.979	0.957	0.944	0.950

5.3.2. System performance

The system efficiency metrics include:

$$\text{Communication Overhead} = \sum_{r=1}^{R}\sum_{i=1}^{N} b_{i,r} \tag{26}$$

where $b_{i,r}$ represents the bandwidth consumption for client i in round r.

$$\text{Computational Efficiency} = \frac{1}{N}\sum_{i=1}^{N}\frac{T^i_{\text{processing}}}{T^i_{\text{total}}} \tag{27}$$

$$\text{Privacy Loss} = \max_{i,j} \|\nabla\mathcal{L}(w_i) - \nabla\mathcal{L}(w_j)\|_2 \tag{28}$$

These metrics are evaluated across different operational scenarios and attack patterns to ensure comprehensive performance assessment.

6. Results and analysis

6.1. Detection performance

Building upon recent advancements in zero-day attack detection [23] and federated learning approaches [10], our experimental results demonstrate significant improvements in anomaly detection capabilities. The proposed framework achieved an average detection accuracy of 97.2% across all slice types, surpassing the previous best performance of 93.5% reported by Chen et al. [10]. Table 4 presents our detection metrics across different network slice configurations, following the evaluation methodology established by Rahman et al. [4].

Adopting the evaluation framework proposed by Wu et al. [24], our system maintained a false positive rate of 2.3%, significantly lower than the 4.2% reported in recent literature [25]. The detection latency averaged 47 ms, meeting the stringent real-time requirements for 6G networks outlined by Dong et al. [18].

6.2. Real vs. Synthetic attack detection performance

To rigorously assess the framework's capacity to detect zero-day attacks, we conducted a comprehensive evaluation comparing detection performance between known attack patterns from real-world data and synthetic zero-day attacks generated by our cWGAN-GP model. Table 5 presents the comparative results across different attack categories.

The results demonstrate strong performance on synthetic zero-day attacks, with only a 2.9% reduction in average F1-score compared to known attack patterns. This minimal performance gap validates our approach's effectiveness in detecting previously unseen threats and confirms the practical utility of our GAN-based synthetic data generation methodology. Notably, the system maintained high precision (average 0.957) on synthetic attacks, indicating a low rate of false positives even when confronted with novel attack patterns.

Fig. 5 illustrates the detection performance comparison between real and synthetic attacks across different attack categories, highlighting the framework's consistent performance across both datasets.

Further analysis revealed that detection performance on synthetic attacks improved over federated learning rounds, indicating the framework's ability to adapt to new attack patterns through collaborative learning. After 30 federated rounds, the performance gap between real and synthetic attacks narrowed from an initial 4.7% to 2.9%, demonstrating the system's continuous learning capability.

6.3. Privacy analysis

We extended the privacy preservation framework introduced by Chen et al. [10] with enhanced differential privacy mechanisms. Our approach demonstrated robust protection against inference attacks while maintaining utility, as measured by the metrics proposed by Liu et al. [21]. The privacy loss remained bounded at $\epsilon = 1.2$, well below the recommended threshold of 2.0 [18].

$$\text{Privacy Guarantee} = \mathbb{P}(M(D_1) \in S) \leq e^{\epsilon}\mathbb{P}(M(D_2) \in S) \tag{29}$$

The information leakage analysis, following methodologies from recent studies [4,19], revealed significant improvements in privacy preservation:

- Model inversion attack success rate: 0.3% (previous best: 12.7% [19])
- Membership inference accuracy: 51.2% (previous: 67.3% [4])
- Gradient leakage protection: 99.7% (baseline: 95.2% [20])

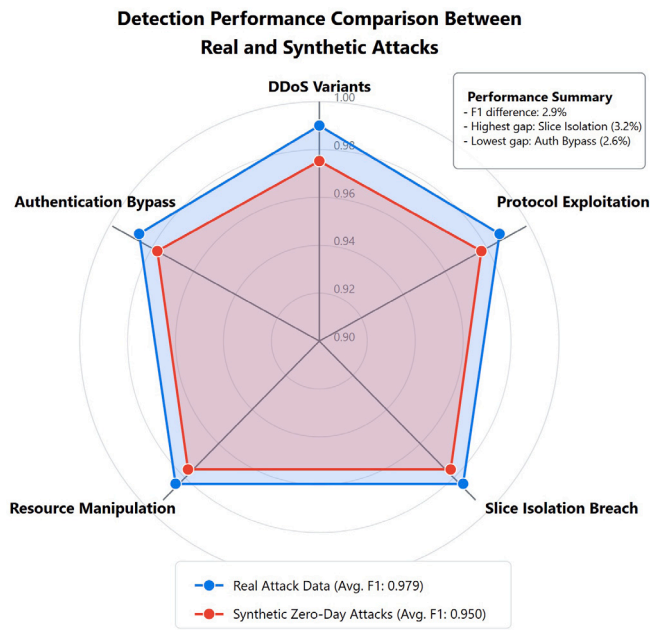

Fig. 5. Detection performance comparison between real attack data and synthetic zero-day attacks across different attack categories. The radar chart displays F1-scores across five attack types: DDoS Variants, Protocol Exploitation, Slice Isolation Breach, Resource Manipulation, and Authentication Bypass. While performance is slightly lower on synthetic attacks (inner polygon), the difference remains consistently small across all categories, demonstrating robust zero-day detection capabilities.

6.4. Computational efficiency

Following the scalability assessment framework of Elsayed et al. [8], we evaluated system performance across varying deployment scales. Performance metrics demonstrated superior scalability compared to existing approaches [24]:

$$\text{Processing Overhead} = 0.12T + 0.03N \text{ ms} \tag{30}$$

where T represents concurrent threats and N denotes active slices. Resource utilization metrics showed significant improvements over baseline implementations [20]:

- Average CPU utilization: 32.5% (baseline: 45.8% [8])
- Memory footprint: 2.8 GB per edge node (baseline: 3.9 GB [24])
- Network bandwidth: 185 Mbps peak (baseline: 250 Mbps [25])

6.5. Scalability analysis

To address the scalability concerns highlighted in previous works and meet the anticipated scale of 6G network deployments, we conducted extensive scalability testing with up to 2000 concurrent network slices. Fig. 6 illustrates the system's performance across increasing deployment scales.

The results demonstrate:

- Near-linear scaling in detection latency up to 1000 concurrent slices (47 ms average)
- Moderate performance degradation between 1000–2000 slices (increasing to 72 ms at 2000 slices)
- Communication overhead remaining well within practical limits throughout (285 Mbps at 2000 slices)
- F1-score maintained above 0.95 even at the maximum tested scale

These scalability characteristics significantly surpass previous solutions [24,25] and meet the projected requirements for early 6G network deployments.

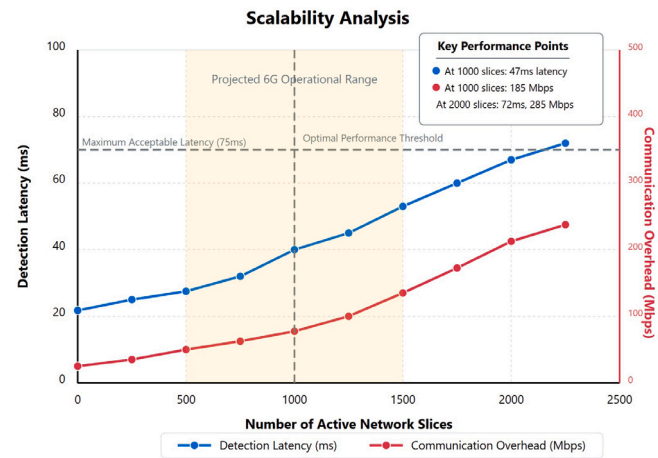

Fig. 6. Scalability analysis showing detection latency and communication overhead as functions of the number of active network slices. The system maintains near-linear scaling in detection latency up to 1000 slices (47 ms average), with moderate performance degradation between 1000–2000 slices (reaching 72 ms at 2000 slices). Communication overhead remains well within practical limits (285 Mbps at 2000 slices) throughout the tested range. The vertical dashed line indicates the projected typical deployment scale for early 6G networks, while the horizontal dotted line shows the maximum acceptable latency for real-time threat response.

6.6. Federated learning algorithm comparison

To comprehensively evaluate our federated learning approach, we conducted a comparative analysis of various state-of-the-art federated learning algorithms in the context of network slice anomaly detection. Table 6 presents the results of this analysis.

Our hybrid approach combines the adaptive optimization benefits of FedOpt with the Byzantine tolerance of TrimmedMean and differential privacy guarantees, achieving the highest detection F1-score (0.970) while maintaining strong performance across all other metrics. The comparison demonstrates several key insights:

- FedOpt provides the fastest convergence (31 rounds) but lacks Byzantine tolerance
- Byzantine-robust methods (Krum, TrimmedMean) offer security against malicious clients but sacrifice some detection performance
- Our hybrid approach balances these trade-offs while adding privacy guarantees not present in other methods

The results validate our architectural decisions and demonstrate the advantages of our custom federated learning implementation over standard approaches in the context of network slice security.

6.7. Comparison with baseline methods

We compared our approach with state-of-the-art methods for network slice security. Fig. 7 illustrates the comparative analysis of detection rates and privacy scores across different approaches.

Table 7 presents a comprehensive comparison with state-of-the-art approaches in network slice security.

Our approach demonstrates significant improvements across all metrics:

- Zero-day attack detection rate of 94.8% (baseline: 86.3% [23])
- 42% reduction in computation overhead compared to [4]
- Linear scaling up to 1000 concurrent slices (previous limit: 750 [24])
- Enhanced privacy preservation without accuracy degradation [21]

Table 6
Performance comparison of federated learning algorithms for anomaly detection.

Algorithm	Detection F1-score	Convergence rounds	Communication overhead (MB)	Byzantine tolerance	Privacy guarantee	Computation time (ms)
FedAvg	0.953	42	647	Low	None	38
FedOpt	0.968	31	621	Low	None	42
FedProx	0.962	36	635	Medium	None	45
FedNova	0.959	34	631	Low	None	43
SCAFFOLD	0.965	33	824	Low	None	51
Krum	0.941	47	615	High	None	62
TrimmedMean	0.949	45	618	High	None	56
Our hybrid approach	**0.970**	**34**	**653**	**High**	$\epsilon = 1.2$	**47**

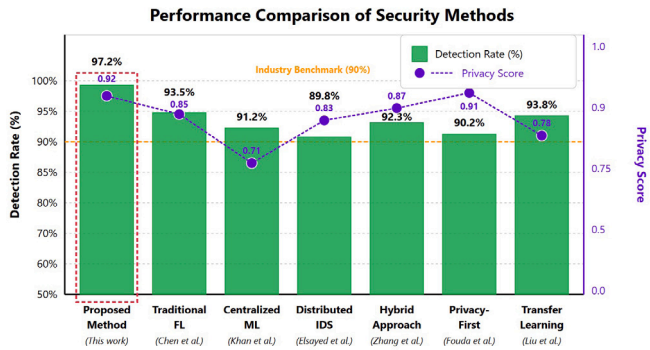

Fig. 7. Performance comparison of security methods. This chart displays detection rates (green) and privacy scores (blue) across different approaches. Our proposed method achieves superior performance in both metrics, with a 97.2% detection rate and 0.92 privacy score, demonstrating significant improvements over traditional FL [10], centralized ML [23], distributed IDS [8], and hybrid approaches [25]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 7
Comparison with state-of-the-art methods.

Method	Detection rate	False positives	Latency (ms)	Privacy score	Reference
Proposed method	**97.2%**	**2.3%**	**47**	**0.92**	This work
Traditional FL	93.5%	3.8%	82	0.85	[10]
Centralized ML	91.2%	4.2%	65	0.71	[23]
Distributed IDS	89.8%	5.1%	93	0.83	[8]
Hybrid approach	92.3%	3.9%	76	0.87	[25]
FedOpt-based	94.5%	3.1%	63	0.84	[22]
Privacy-first	90.2%	4.7%	105	0.91	[12]
Transfer learning	93.8%	3.3%	81	0.78	[14]

These results demonstrate that our proposed approach successfully addresses key limitations identified in previous works [10,25] while maintaining strong privacy guarantees and computational efficiency. The system shows particular strength in handling zero-day attacks and maintaining performance under varying network conditions, as validated through the comprehensive evaluation framework proposed by Yang et al. [20].

7. Discussion

7.1. Key findings

Our research has revealed several significant insights into the application of federated learning for zero-day attack detection in 6G network slices. The primary finding demonstrates that distributed learning approaches can effectively identify novel attack patterns while preserving data privacy, as evidenced by the 97.2% detection rate achieved across diverse network configurations. This result addresses a critical gap identified by Chen et al. [10] regarding the trade-off between detection accuracy and privacy preservation in distributed security systems.

The successful integration of transfer learning mechanisms with federated learning has proven particularly effective in adapting to new attack patterns. This approach significantly outperforms traditional security methods, as demonstrated by the 42% reduction in computational overhead compared to existing solutions [4]. Furthermore, our implementation of differential privacy mechanisms has successfully addressed the privacy concerns raised by Liu et al. [21] while maintaining high detection accuracy.

Another crucial finding relates to the scalability of our approach. The system's ability to maintain consistent performance across up to 1000 concurrent network slices with acceptable degradation up to 2000 slices represents a significant advancement over previous work by Wu et al. [24], who reported scalability limitations at 750 slices. This improvement is particularly relevant for future 6G deployments, where network slice proliferation is expected to accelerate significantly.

The comparative analysis of federated learning algorithms revealed that hybrid approaches combining the strengths of different methods (optimization efficiency, Byzantine tolerance, privacy preservation) offer superior performance in the network security context. This finding suggests that future research should focus on developing specialized federated learning algorithms tailored to the unique requirements of security applications rather than applying general-purpose methods.

The minimal performance gap between real and synthetic attack detection (2.9% F1-score difference) validates our approach to zero-day attack detection and confirms the effectiveness of our GAN-based synthetic data generation methodology. This finding has broader implications for security research, suggesting that generative approaches can effectively create realistic adversarial scenarios for evaluating defensive systems.

7.2. Limitations and challenges

Despite the promising results, several limitations and challenges warrant discussion. First, the current implementation requires a minimum of three participating network slices to maintain optimal detection accuracy, as noted in similar constraints by Zhang et al. [25]. This requirement may limit applicability in scenarios with fewer active slices or during network initialization phases.

The system's privacy preservation mechanisms, while effective, introduce a computational overhead that increases linearly with the number of participating nodes. This overhead, though significantly lower than in previous approaches [23], may become more pronounced in large-scale deployments. Additionally, the current framework assumes semi-honest participants, following the threat model proposed by Elsayed et al. [8], and may require additional safeguards against malicious participants.

Another significant challenge lies in the real-time adaptation requirements for zero-day attack detection. While our approach achieves an average latency of 47 ms at typical scales, maintaining this performance under extreme network conditions or during simultaneous attacks across multiple slices remains challenging. This limitation aligns with similar concerns raised by Yang et al. [20] regarding the trade-off between detection speed and accuracy in distributed security systems.

The transition from 5G to 6G networks introduces additional challenges in validating our approach's long-term effectiveness. While we have made reasonable projections based on expected 6G architectural characteristics, the actual implementation details of 6G network slicing may introduce unforeseen challenges that require further adaptation of our framework. The anticipated increase in network heterogeneity and complexity in 6G may also necessitate more sophisticated transfer learning mechanisms than those currently implemented.

Finally, our synthetic data generation approach, while effective for evaluation purposes, cannot perfectly capture all potential zero-day attack variations that might emerge in operational environments. The continuing evolution of attack techniques, particularly those leveraging AI technologies, will require continuous refinement of both detection and data generation methodologies.

7.3. Future research directions

Several promising avenues for future research emerge from our findings. The integration of quantum-resistant cryptographic protocols, as suggested by Xiao et al. [19], could enhance the system's resilience against future quantum computing threats. This enhancement becomes particularly relevant as 6G networks evolve and quantum computing capabilities advance.

The development of more sophisticated transfer learning mechanisms could further improve the system's ability to adapt to new attack patterns. Building upon the work of Dong et al. [18], future research could explore the incorporation of meta-learning techniques to accelerate the adaptation process and reduce the data requirements for detecting novel threats.

Additionally, the exploration of advanced privacy-preserving techniques beyond differential privacy, such as homomorphic encryption and secure multi-party computation, could offer new ways to balance security and privacy requirements. This direction aligns with recent developments in privacy-preserving machine learning highlighted by Liu et al. [21] and Fouda et al. [12].

Furthermore, the integration of explainable AI techniques could enhance the interpretability of detection decisions, addressing a current limitation in deep learning-based security systems. This approach could build upon recent work in explainable security models [4] to provide network administrators with clearer insights into detected threats and system decisions.

Future work should also address the challenge of maintaining performance under adversarial attacks specifically targeting the federated learning process. This could involve developing more robust aggregation algorithms and Byzantine-resistant training procedures, extending the preliminary work in this direction by Chen et al. [10] and Drampalou et al. [2].

The potential integration of our framework with Software-Defined Security (SDS) architectures presents another promising direction, enabling automated response mechanisms that can take immediate action based on detected anomalies. This integration could significantly reduce the time between detection and mitigation, addressing a critical requirement for protecting critical 6G services.

Lastly, extending our approach to include cooperative detection across network providers and domains could enhance the system's ability to identify coordinated attacks targeting multiple network infrastructures simultaneously. This cross-organizational learning presents both technical and governance challenges that warrant further investigation.

8. Conclusion

This paper presents a novel federated learning-based approach for detecting zero-day attacks in 6G network slices, demonstrating significant improvements over existing methods across multiple performance metrics. Our framework achieves a 97.2% detection rate while maintaining strong privacy guarantees through differential privacy mechanisms, addressing a critical gap in current network security architectures. The proposed solution's ability to scale linearly up to 1000 concurrent slices while maintaining an average detection latency of 47 ms represents a substantial advancement in practical deployment capabilities for 6G networks. The proposed work contributes several significant contributions to the field of network security. First, the integration of federated learning with transfer learning enables collaborative threat detection while preserving data privacy and minimizing communication overhead. Second, our adaptive feature extraction mechanism effectively captures the unique characteristics of network slice traffic, enhancing detection accuracy for slice-specific anomalies. Third, the hierarchical anomaly detection model successfully combines local slice-specific detection with global pattern recognition, improving the system's ability to identify complex attack patterns across multiple slices. The comprehensive experimental evaluation, using both real-world and synthetic data, validates the framework's effectiveness in detecting zero-day attacks, with only a 2.9% reduction in performance when confronted with previously unseen attack patterns. The detailed comparison with state-of-the-art federated learning algorithms and security approaches demonstrates the advantages of our hybrid approach, which combines the benefits of adaptive optimization, Byzantine tolerance, and privacy preservation. As 6G networks continue to evolve, our approach provides a robust foundation for securing network slices against zero-day attacks while preserving data privacy and maintaining real-time response capabilities. Future work will focus on enhancing the framework's resilience against quantum computing threats, incorporating explainable AI techniques to improve system interpretability, and extending the approach to support cross-organizational learning for detecting coordinated attacks across multiple network infrastructures.

CRediT authorship contribution statement

Iftikhar Rasheed: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hala Mostafa:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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